J. Felipe GONZALEZ

Research and Teaching Statement

Research Statement

My research lies at the intersection of Human–Computer Interaction (HCI), interactive systems, and computational tools for creation. I study how computational systems can better support the ways people design, create, and collaborate across digital and physical environments. My goal is to develop interactive systems that are not only technically capable, but also understandable, learnable, and effective for the people who use them.

I am particularly interested in contexts where interaction is technically sophisticated and cognitively demanding: working with 3D models, moving between visual and computational representations, interacting in immersive environments, and using tools that must support both precision and exploration. Across these contexts, I investigate questions such as: How can interfaces reduce friction in complex tasks? How can interaction techniques better support understanding and control? How can computational assistance be integrated in ways that improve human performance without obscuring system behaviour?

My work has developed through two complementary directions. The first is the design of creator tools for 3D modeling, programming-based CAD, and digital fabrication. The second is interaction in immersive systems such as VR/XR. I see these not as isolated topics, but as related settings for studying a broader problem: how interactive systems can better support creation and action in increasingly rich digital environments. This makes my work a natural fit for a research environment in Graphics and Interaction, where the design of interfaces, visual systems, and new forms of interactive media are closely connected.

Current Work

The first major direction of my research emerged from my doctoral work on programming-based 3D CAD for digital personal fabrication. In this context, users define geometric models through code rather than by direct manipulation. This creates distinctive HCI challenges: users must understand the relation between abstract textual descriptions and visual geometry, verify correctness, reason parametrically, and anticipate fabrication problems. My work studied these usability and learnability challenges and showed that even expert users encounter significant friction when moving between code and visual structure.

Building on this, I introduced bidirectional techniques that connect code and direct manipulation. Rather than treating programming and interaction as separate modes, this work allows users to manipulate geometric elements visually while preserving consistency with the underlying code. This line of research contributes to the design of creator tools that better support modeling, reuse, iteration, and understanding. More broadly, it speaks to a general HCI problem: how to design interfaces for computationally expressive tools without overwhelming users.

The second major direction of my research focuses on interaction in immersive systems. In VR/XR, I have studied haptic feedback, target selection, input techniques, avatar representation, and collaborative immersive environments. This work examines how different representations, feedback mechanisms, and input modalities affect user performance and experience in 3D spatial tasks. More broadly, it contributes to understanding how people act in systems that move beyond conventional desktop interaction and instead rely on embodiment, multimodal feedback, and spatial coordination.

These two directions are connected by a common research question: how can interactive systems better support human understanding and action in complex digital environments? In both creator tools and immersive systems, users must coordinate perception, intention, and system behaviour under conditions that are often cognitively demanding. This shared foundation is what connects my past work to my future agenda.

Short-Term Research Direction

The first short-term direction is to continue developing interactive systems for 3D design, programming-based CAD, and digital fabrication. I am especially interested in tools that better support modeling, iteration, and communication between visual and computational representations. This includes reducing barriers to entry for newcomers, improving workflows for modifying existing models, and helping users understand how design choices propagate through parametric systems. A central goal is to make technically powerful tools more approachable without sacrificing expressiveness.

The second short-term direction is to continue exploring how immersive technologies can be introduced into the construction of 3D models. While most 3D modeling workflows still rely on desktop-based interfaces, immersive environments offer opportunities for more direct spatial interaction, richer visual feedback, and new ways of understanding geometry and scale. In the short term, I am interested in studying how VR/XR can support tasks such as model exploration, editing, validation, and communication during the design process. More broadly, this direction would allow me to investigate how immersive interaction can complement existing modeling workflows and open new possibilities for creating and refining 3D content.

Across both of these directions, I am increasingly interested in selective and well-scoped uses of AI. In the short term, this means exploring how AI can provide modeling support, feedback, guidance, and interaction assistance while preserving usability, transparency, and meaningful user control. I am particularly interested in improving the way AI systems communicate with users so that their assistance becomes not only more effective, but also easier to understand and evaluate.

Medium-Term Research Direction

In the medium term, I aim to expand from individual interaction techniques toward broader systems in which users coordinate with multiple representations, devices, and collaborators.

One part of this agenda is the development of richer creator workflows that connect 3D design, interactive visualization, and fabrication. I am interested in systems that help users move more fluidly between code, direct manipulation, simulation, and physical outcomes. This would extend my previous work on programming-based CAD toward broader environments for computational design and interactive graphics.

A second part is the study of immersive systems for collaborative work. I would like to investigate how XR can support co-located and distributed collaboration in technical and creative tasks, including how shared representations, avatars, and multimodal interaction affect coordination and performance. This connects directly with my current work on collaborative immersive systems and would allow me to build stronger bridges between HCI, graphics, and spatial computing.

A third part is the study of AI-supported interaction in these systems. I am interested in how intelligent assistance can be integrated into creator tools and immersive environments without undermining human understanding or agency. This includes questions about feedback, trust, interpretability, and how systems should present suggestions or adapt to users in ways that remain intelligible and useful.

Long-Term Research Direction

In the long term, my goal is to build an integrated research program on interactive systems for creation, collaboration, and computational work, with impact across HCI, Graphics and Interaction, and related areas of computer science.

One long-term objective is to develop generalizable design principles for tools that combine visual interaction, computational structure, and intelligent assistance. Rather than treating each application as isolated, I want to contribute broader knowledge about how interactive systems should support human action in complex design and decision-making tasks.

A second long-term objective is to contribute to future environments for creation and digital work. I am interested in systems where design, visualization, immersive interaction, and computational assistance

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are tightly connected, allowing users to move fluidly across media, devices, and levels of abstraction. This could apply to domains such as digital fabrication, 3D content creation, technical workflows, and collaborative design.

A third long-term objective is to ensure that such systems remain accessible and human-centered. I want to investigate how advanced tools can support a wider range of users, including novices and people with diverse backgrounds and abilities. In this sense, accessibility, learnability, and transparency are not side concerns, but central design goals.

Overall, I aim to build a research program that contributes both technically and conceptually: technically, by creating new interactive systems and tools; conceptually, by improving our understanding of how graphics, interaction, immersion, and intelligent assistance can be combined in ways that better support human goals.

Teaching Statement

My teaching philosophy is centered on clarity, reasoning, and progressive skill building. I want students to understand why methods work, not only how to reproduce them. In my experience, students learn more effectively when concepts are introduced incrementally, connected to prior knowledge, and reinforced through examples, guided exercises, and project-based work. I also believe that mistakes are an important part of learning, especially in technical areas such as programming and systems design, where debugging and revision are essential intellectual practices.

More specifically, my teaching is guided by four principles. **First**, I focus on reasoning rather than mechanical procedures. I want students to move beyond pattern imitation and develop robust mental models of the concepts they are learning. **Second**, I build understanding incrementally, making explicit how new topics connect to previous ones so that students can organize knowledge as a coherent structure rather than as isolated units. **Third**, I treat mistakes as opportunities for learning. In technical disciplines, errors are not simply failures to avoid, but opportunities to analyze assumptions, revise understanding, and develop stronger habits of thought. **Fourth**, I recognize that students arrive with different backgrounds and preparation levels, so I try to provide multiple entry points into the material through visual explanations, analogies, worked examples, and hands-on activities. My goal is not to lower standards, but to help a broader range of students reach them.

My teaching experience spans more than 1,500 hours across mathematics, programming, computer science, and engineering. I have worked with students at different stages, from early undergraduate courses to graduate-level teaching. Last year, I served as an instructor for a graduate course on Data Interaction Techniques. These experiences have made me comfortable teaching both foundational and advanced topics, and adapting explanations to students with different backgrounds and levels of preparation.

At Instituto Superior Tecnico, I would be glad to contribute to teaching in a broad but coherent set of areas. At the undergraduate level, I would be comfortable teaching programming-related courses, object-oriented programming, introductory software design, project-based computing courses, and courses related to systems and interaction. I could also contribute to courses in HCI, graphics, visualization, VR/XR, and related interactive technologies. At the graduate level, I would be particularly interested in seminars and project-based courses related to interactive systems, immersive technologies, creator tools, design and evaluation methods, and computational media.

My research would directly enrich this teaching. In programming and systems courses, I can bring examples from computational design, prototyping, and digital fabrication. In graphics, interaction, and XR courses, I can integrate examples involving immersive systems, prototyping, and empirical evaluation. More broadly, I want students to leave my courses not only with technical competence, but also with the ability to reason critically about interfaces, tools, users, and the broader implications of computing technologies.

Teaching, for me, is closely tied to mentoring. During my Ph.D. and postdoctoral work, I have supervised and co-supervised projects with Master's and Ph.D. students, and I have also acted as an informal mentor in research on XR interaction, study design, analysis, and academic writing. This mentoring has contributed to peer-reviewed publications and to the development of student-led projects. My advising style combines structure with flexibility: I aim to help students define clear goals, break down complex problems into manageable milestones, and develop strong habits in reading, implementation, documentation, and critical reflection.

Overall, I aim to contribute to Instituto Superior Tecnico with a combined profile in research, teaching, and mentoring. My goal is to build a research program at the intersection of HCI, 3D design, immersive systems, and emerging AI-supported interaction, while contributing to teaching in programming, systems, graphics, interaction, visualization, and related interactive technologies.

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